ANSWER KEY & MARKING SCHEME SET A

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 50 N/m²	Pressure = force ÷ area
2	**(d)** force per unit area	P = F/A
3	**(a)** in all directions	At bottom and on sides
4	**(b)** atmosphere	Contains nitrogen and oxygen
5	**(b)** 100 Pa	Practical unit
6	**(c)** from sea to land during the day	Land heats faster
7	**(b)** keep doors and windows open (reducing pressure difference)	Reduces roof lifting
8	**(c)** positive charges	Lighter ice particles
9	**(b)** warm ocean waters	Warm moist air
10	**(b)** 270 km/h	As given in the chapter

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force per unit area (1). P = F/A (1).	1+1
12	Pascal (1). 1 Pa = 1 N/m² (1).	1+1
13	Larger area reduces pressure (1). Same weight is spread out (1).	1+1
14	Liquids exert pressure in all directions (1). Particles push on every surface (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	P = 300 ÷ 1.5 = 200 N/m² (1½). New P = 300 ÷ 3 = 100 N/m² (1½).	1½+1½
16	Pointed end has small area (1). Same force on small area gives large pressure (1). Pressure can penetrate wood (1).	1+1+1
17	Liquid pressure increases with depth (1½). Bottom must withstand large horizontal pressure (1½).	1½+1½
	OR: Height of water column is greater for lower floor (1½). Greater height gives greater pressure (1½).	1½+1½
18	Area 225 cm² (1). Force about 2250 N (1). Equal to weight of 225 kg (1).	1+1+1
	OR: Press a sucker on smooth wall (1). It sticks since air is pushed out (1). Outside pressure is greater than inside (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pressure = F/A (1). More force gives more pressure (1). Less area gives more pressure e.g. knife (1½). More area less pressure e.g. broad strap (1½).	1+1+1½+1½
	OR: Acts in all directions (1). Increases with height of column (1). Dam base broad (1½). Water tank at height (1½).	1+1+1½+1½
20	Pressure by air (1). Rubber sucker sticking (2). Balloon expanding or bottle crushing (2).	1+2+2
	OR: High to low pressure principle (1). Day: sea breeze (1½). Night: land breeze (1½). Diagram labelled (1).	1+1½+1½+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 100 N/m²	Pressure = force ÷ area
2	**(d)** pascal (N/m²)	Also called Pa
3	**(b)** height of the liquid column	Greater depth gives greater pressure
4	**(a)** atmospheric pressure	Acts on all objects
5	**(b)** high pressure to low pressure	Air moves until pressure equalises
6	**(c)** from land to sea at night	Sea is warmer at night
7	**(d)** rises, creating low pressure	Cooler air rushes in
8	**(b)** negative charges	Heavier water droplets
9	**(d)** lowest pressure and calm wind	Calm centre
10	**(b)** West Bengal and nearby regions	Pre-monsoon storm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Greater height gives greater pressure (1). Water flows with force from taps (1).	1+1
12	Pressure exerted by the air around us (1). It acts in all directions on all objects (1).	1+1
13	Pressure inside our body is equal to outside (1). So the pressures balance (1).	1+1
14	Air is pushed out, reducing pressure inside (1). Outside air pressure is higher and presses it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Day: land warms, air rises, wind sea to land (1½). Night: sea warmer, low pressure over sea, wind land to sea (1½).	1½+1½
16	Wind passes through the house (1). Pressure difference between inside and outside is reduced (1). Roof is less likely to be blown away (1).	1+1+1
17	Land heated, warm moist air rises (1). Low pressure forms; cooler air rushes in (1). Water vapour condenses forming clouds, rain and strong winds (1).	1+1+1
	OR: Ice particles and water droplets rub in strong winds (1). Positive charges collect at top, negative at bottom (1). Ground becomes positive by induction (1).	1+1+1
18	Stay indoors or in a car (1). Avoid tall trees and metal objects (1). Crouch low in open area, do not lie flat (1).	1+1+1
	OR: Metal rod fixed along wall with pointed top above building (1). Lower end buried deep in ground (1). It gives charges an easy path to the ground (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	High speed means low pressure (1). Balloon activity (2). Roof example (2).	1+2+2
	OR: Warm air rises, low pressure (1). Cool air rushes in (1). Condensation forms clouds (1). Charges produced in clouds and lightning flashes (1). Thunder follows (1).	1+1+1+1+1
20	Rubbing ice and water charges clouds (1). Top positive, bottom negative (1). Air insulation breaks, flash (1). Safety: stay low, avoid tall objects (1). Avoid water and metals (1).	1+1+1+1+1
	OR: Diagram (1). Metal rod above the building (1). Other end in ground (1). Provides easy path (1). Charges flow harmlessly to ground (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.